LAXMI PUBLIC SCHOOL

Kailashpuri, Mughalsrai,, Chandauli

SUMMER VACATION HOMEWORK 2026-27

CLASS- XII SCIENCE

[विषय – हिन्दी]

परियोजनाकार्य :-

1. पत्रकारीय लेखन क्या है? पत्रकारीय लेखन का संबंध व उद्देश्य, व पत्रकार के प्रकारों का सचित्र वर्ण न करें।
2. समाचार लेखन के छहककार क्या हैं? और समाचार कैसे लिखा जाता है?
3. संपादकीय लेखन क्या है? संपादकीय लेखन का प्रारूप क्या है? एवं संपादक की क्या भूमिका होती है? सचित्र लिखें।
4. पाँच दिन का समाचार पत्र लिखिए।
5. भक्तिन के जीवन के संघर्षों व चारों"परिच्छेदों का सचित्र वर्णन करें।
6. 'जूझ' पाठकेआधार पर उसके संघर्षों का सचित्र वर्णन करें।
7. 'आत्मपरिचय' पाठकेआधार पर हरिवंशरायबच्चन का आत्मपरिचय सचित्र लिखिए।

[English]

1. Solve previous year Board Question Paper in your class work copy. Only Reading and writing section. (year 2022,2023,2024)

2. Choose a novel, play, poem short story and analyze its themes, characters, symbolism or language in a project file.

[Physics]

Q.1. To write activity on -

(a) To assemble a household circuit comprising three bulbs, three (on/off) Switches, a f use and a power source.

(b) To assemble the components of a given electrical circuit.
(c) To draw the diagram of a given open circuit Comprising at least a battery, resistor /rheostate, ammeter and voltmeter.

(d) To identity a diode, an LED, a transistor, an IC. a resistor and a capacitor from a mixed collection of such items.

(e) Use of multimeter to see the unidirectional flow of current in case of a diode and an LED and cheek whether given electronic Component (eg: diode) is in working order

(f) To observe diffraction of light due to a thin slit.

Q.2. To solve five years Board Question Paper.

(a) Electric Fields and charges (b) Electric Potential and capacitance

Investigatory project for AISSC Examination 2026-27

1. Water level indicator
2. Li-Fi Audio system
3. Wireless Mobile charger
4. Security alarm System

5. Toll tax device
6. Automatic Night bulb
7. Clapping Switch
8. Rectifier (Half wave / full wave)
9. P-N junction diode forward and Reverse bias
10. Windmill Night lamp

[Chemistry]

1. Complete your investigatory project which you have been already allotted (except the observation part if you haven't performed your project).

PROJECT FILE SHOULD CONTAIN PAGES IN FOLLOWING ORDER:

a, CERTIFICATE
b. ACKNOWLEDGEMENT
c. AIM OF PROJECT
d. INTRODUCTION e, THEORY
f. APPARATUS REQUIRED
g. PROCEDURE
h. OBSERVATION
i. CONCLUSION . PRECAUTION k, BIBLIOGRAPHY

** STRICTLY ADHERE TO ABOVE MENTIONED ORDER

2. Make a list of all the named reactions in chapter 'Haloalkanes & Haloarenes' and 'Alcohols, Phenols & Ether' write them in your assignment register.

3. Write all the reaction mechanism in chapter 'Haloalkanes & Haloarenes' and 'Alcohols, Phenols & Ether' in your assignment register.

4. List five distinguish test with example studied till yet.

INVESTIGATORY PROJECTS

Roll,no 1to5_Project 1.
Study the presence of oxalate ions in guava fruit at different stages of ripening

Roll, no 6to10- Project 2.
To study of quantity of casein present in different samples of milk.

Roll, no 11to15 Project 3.
Preparation of soyabean milk and its comparison with natural milk with respect to curd formation, effect of temperature, etc.

Roll, no 16to20- Project 4.
To study the effect of potassium bisulphate as food preservative under various conditions (temperature, concentration, time, etc.)

Roll, no 21to25Project 5.
To study of digestion of starch by salivary amylase and effect of pHand temperature on it.
Roll, no 26to30Project 6.

A comparative study of the rate of fermentation of following materials: wheat flour, gram flour, potato juice, carrot juice, etc.

Roll,no 31to35Project 7.

Extraction of essential oils present in saunf (aniseed), again (carum), illaichi (cardamom).

Roll,no 36to40Project 8.

To study the common food adulterants in fat, oil, butter, sugar, turmeric powder, chilli powder and pepper.

Answer the following questions-

SOLUTION

1	State Henry' slaw correlating the pressure of a gas and its solubility in a solvent and mention Two applications of the law.
2	Calculate the temperature at which a solution containing 54g of glucose,(C6H12O6) in 250g of water will freeze. (Kffor water =1.86 K mol kg $^{-1}$)
3	State Raoult's law for solutions of volatile liquids.Taking suitable examples explain the Meaning of positive and negative deviations from Raoult's law. **OR** Define the term osmotic pressure. Describe how the molecular mass of a substance canbe determined by a method based on measurement of osmotic pressure?
4	Define osmotic pressure. How is it that measurement of osmotic pressures is more widely used for determining molar masses of macromolecules than the rise in boiling point or fall in freezing point of their solutions? OR Derive an equation to express that relative lowering of vapour pressure for a solution is equal to the mole fraction of the solute in it when the solvent alone is volatile.
5	Differentiate between molality and molarity of a solution. What is the effect of change in temperature of a solution on its molality and molarity?
6	(a)Define the following terms:(i) Mole fraction (ii) Van't Hoff factor (b) 100 mg of a protein is dissolved in enough water to make 100 mL of a solution. If this solution has an osmotic pressure13.3 mm Hg at 25° C, what is the molar mass of protein?(R =0.0821 Latmmol^{-1}K^{-1}and760mmHg=1atm.) OR What is meant by: (i) Colligative properties (ii) Molality of a solution. (b)Whatconcentrationofnitrogenshouldbepresentinaglassofwateratroomtemperature? Assumeatemperatureof25°C,totalpressureof1atmosphereandmolefractionofnitrogen in air of 0.78. [KH for nitrogen = 8.42 × 10^{-7}M/mm Hg]
7	Calculate the freezing point depression for 0.0711 m aqueous solution of sodium sulphate (Na2 SO4) ,if it is completely ionised in solution. If this solution actually freezes at–0.320°C,what is the value of Van't Hoff factor for it at the freezing point? (Kf for water is 1.86°C mol–1)

8	What is ‘reverse osmosis’?
9	Non-ideal solutions exhibit either positive or negative deviations from Raoult' slaw. What are these deviations and why are they caused? Explain with one example for each type.

10 A solution prepared by dissolving 1.25 g of oil of winter green (methyl salicylate) in 99.0 g of benzene has a boiling point of 80.31°C. Determine the molar mass of this compound. (B.P. of pure Benzene = 80.10°C and Kb for benzene = 2.53°C kg mol–1)

11 A solution of glycerol ($C_3H_8O_3$;molar mass=92gmol-1)in water was prepared by dissolving some glycerol in 500 g of water.This solution has a boiling point of 100.42°C.What mass of glycerol was dissolved to make this solution? Kb for water =

12 Define the terms, ‘osmosis’ and ‘osmotic pressure’. What is the advantage of using osmotic pressure as compared to other colligative properties for the determination of molar masses of solutes in solutions.

13 What mass of NaCl (molar mass = 58.5 g mol–1) must be dissolved in 65 g of water to lower the freezing point by 7.5°C? The freezing point depression constant , Kf, for water is 1.86 K kg mol–1. Assume van’t Hoff factor for NaCl is 1.87.

14 What mass of ethylene glycol (molar mass =62.0 g mol–1) must be added to 5.50 kg of water to lower the freezing point of water from 0°C to – 10.0°C? (Kf for water = 1.86 K kg mol–1)

15 15 g of an unknown molecular substance was dissolved in 450 g of water. The resulting solution freezes at –0.34°C.What is the molar mass of the substance? (Kf for water=1.86 K kg mol–1).

16 (a)Differentiate between molarity and molality for a solution. How does a change in temperature influence their values?

(b) Calculate the freezing point of an aqueous solution containing 10.50 g of $MgBr_2$ in 200 g of water. (Molar mass of $MgBr_2$= 184 g) (Kf for water = 1.86 K kg mol–1)

OR

(a) Define the terms osmosis and osmotic pressure. Is the osmotic pressure of a solution a colligative property? Explain.

(b) Calculate the boiling point of a solution prepared by adding 15.00gofNaClto250.00g of water. (Kb for water = 0.512 kg mol–1), (Molar mass of NaCl = 58.44 g)

17 (a)State the following: (i) Henry’s law about partial pressure of a gas in a mixture. (ii)

Raoult’ s law in its general form in reference to solutions.

(b) A solution prepared by dissolving 8.95 mg of a gene fragment in 35.0 mL of water has an osmotic pressure of 0.335 torr at 25°C. Assuming the gene fragment is a nonelectrolyte, determine its molar mass.

OR

(a) Differentiate between molarity and molality in asolution. What is the effect of temperature change on molarity and molality in a solution?

(b) What would be the molar mass of a compound if 6.21g of it dissolved in 24.0 g of

	chloroform form a solution that has a boiling point of 68.04°C. The boiling point of pure chloroform is 61.7°C and the boiling point elevation constant, Kb for chloroform is 3.63°C/m.
18	A0.561 m solution of an unknown electrolyte depresses the freezing point of water by 2.93°C.What is Van't Hoff factor for thise lectrolyte? The freezing point depression Constant (Kf) for water is 1.86°C kg l–1.
19	A 1.00 molal aqueous solution of trichloroacetic acid (CCl3COOH) is heated to its boiling point.The solution has the boiling point of 100.18°C. Determine the van't Hoff factor for
	trichloroacetic acid. (Kb for water = 0.512 K kg mol–1) OR Define the following terms: (i) Mole fraction (ii) Isotonic solutions (iii) Van't Hoff factor (iv) Ideal solution
20	Calculate the amount of KCl which must be added to 1 kg of water so that the freezing point is depressed by 2 K. (Kf for water = 1.86 K kg mol–1)

[Biology]

- Write the experiments in your lab manual.
1. Study the plant population density by quadrat method.
2. Study the plant population frequency by quadrat method.
3. Prepare a temporary mount of onion root tip to study mitosis.
4. Isolate DNA from available plant material such as spinach, green pea seeds, papaya etc.
5. Flowers adapted to pollination by different agencies (wind, insect, birds).
6. Pollen germination on stigma through a permanent slide.

- Prepare an Investigatory file on any one of the following topics. Make sure your file should be well decorated and contains correct contents.
1. AIDS
2. Cancer
3. Reproductive Health
4. Menstrual cycle
5. Ecosystem
6. Application of Biotechnology
7. Recombinant DNA technology
8. Pollination
9. Impact of pollution
9. Antibiotic resistance
10. DNA fingerprinting
11. Study of genetic disorder
12. Infertility
13. Mendelian Inheritance
14. Human Health & Diseases
15. Microbes in Human welfare
16. PCR & its applications
17. Drug Addiction

- Solve PYQs from year 2022-26 of chapter 1-3 in your Biology Notebook & also mention the year.

[Mathematics]

1. Create a stick file on the topic allotted to the student in the class.
2. Prepare a formula chart of inverse trigonometry function, trigonometry transformation, differentiation.
3. Solve previous 5 year questions of chapter taught in the class.

[Physical Education]

Activity:-

Answer the following Questions

Q-1 How yoga can benefit the Asthma and Hypertension patient? Explain

Q-2 Highlight the yogic techniques to cure Obesity

Q-3 Explain how Anulom-Vilom is different from Kapal bhati.

Q-4 What do you mean by Diabetes? Explain.

Solve PYQs from year 2022-26 of Chapter 1-3 in your Physical Education Notebook & also mention the year.

[Painting]

1. To solve 5 years Board Question Paper.
2. Draw a two drawing sheets with colours & shade it.

[Hindustani Music Vocal]

1. Practical file:- Rag Bhairav- Description and notation, Taal, Ektal, Jhaptal,Rupak taal, Teental,Dadra, Kaharwa